

More than two groups: ANOVA

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ANOVA

(**AN**alysis **Of** **V**ariance)

- ***Idea:*** For two or more groups, test difference between means, for quantitative normally distributed variables.

One-Way Analysis of Variance (单因素方差分析)

- Assumptions:
 - Normally distributed outcome
 - Equal variances between the groups
 - Groups are independent

Hypothesis of One-Way ANOVA

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots$$

H_1 : Not all of the population means are the same

Why ANOVA?

- If we have three (or more) groups to compare:
 - Multiple comparison would increase type I error.
 - Can we do the pairwise differences “all at once.”

The “F-test”

Is the difference in the means of the groups more than background noise (=variability within groups)?

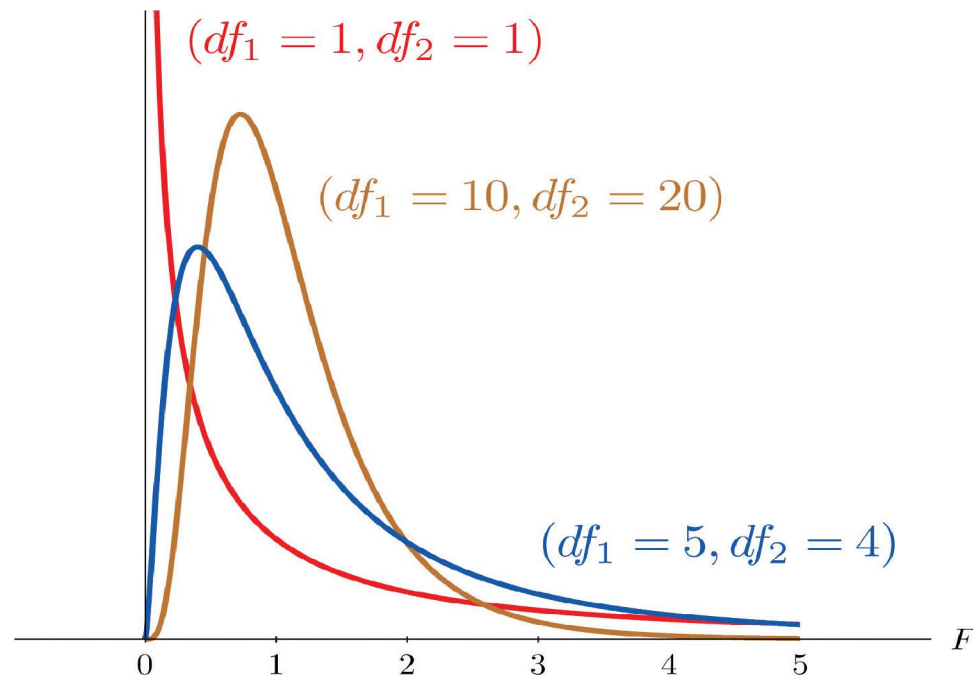
Summarizes the mean differences between all groups at once.

$$F = \frac{\text{Variability between groups}}{\text{Variability within groups}}$$

Analogous to t-test.

The F-distribution

- The F-distribution is a continuous probability distribution that depends on two parameters n and m (numerator and denominator degrees of freedom, respectively):



The F-distribution

- A ratio of variances follows an F-distribution:

$$\frac{\sigma_{between}^2}{\sigma_{within}^2} \sim F_{n,m}$$

- The F-test tests the hypothesis that two variances are equal.
- F will be close to 1 if sample variances are equal.

How to calculate ANOVA's by hand...

Treatment 1	Treatment 2	Treatment 3	Treatment 4
y_{11}	y_{21}	y_{31}	y_{41}
y_{12}	y_{22}	y_{32}	y_{42}
y_{13}	y_{23}	y_{33}	y_{43}
y_{14}	y_{24}	y_{34}	y_{44}
y_{15}	y_{25}	y_{35}	y_{45}
y_{16}	y_{26}	y_{36}	y_{46}
y_{17}	y_{27}	y_{37}	y_{47}
y_{18}	y_{28}	y_{38}	y_{48}
y_{19}	y_{29}	y_{39}	y_{49}
y_{110}	y_{210}	y_{310}	y_{410}

$n=10$ obs./group

$k=4$ groups

$$\bar{y}_{1\bullet} = \frac{\sum_{j=1}^{10} y_{1j}}{10}$$

$$\bar{y}_{2\bullet} = \frac{\sum_{j=1}^{10} y_{2j}}{10}$$

$$\bar{y}_{3\bullet} = \frac{\sum_{j=1}^{10} y_{3j}}{10}$$

$$\bar{y}_{4\bullet} = \frac{\sum_{j=1}^{10} y_{4j}}{10}$$

$$\frac{\sum_{j=1}^{10} (y_{1j} - \bar{y}_{1\bullet})^2}{10 - 1}$$

$$\frac{\sum_{j=1}^{10} (y_{2j} - \bar{y}_{2\bullet})^2}{10 - 1}$$

$$\frac{\sum_{j=1}^{10} (y_{3j} - \bar{y}_{3\bullet})^2}{10 - 1}$$

$$\frac{\sum_{j=1}^{10} (y_{4j} - \bar{y}_{4\bullet})^2}{10 - 1}$$

The group means

The (within)
group variances

Sum of Squares Within (SSW)

$$\sum_{j=1}^{10} (y_{1j} - \bar{y}_{1\bullet})^2 + \sum_{j=1}^{10} (y_{2j} - \bar{y}_{2\bullet})^2 + \sum_{j=3}^{10} (y_{3j} - \bar{y}_{3\bullet})^2 + \sum_{j=1}^{10} (y_{4j} - \bar{y}_{4\bullet})^2$$

$$= \sum_{i=1}^4 \sum_{j=1}^{10} (y_{ij} - \bar{y}_{i\bullet})^2$$

Sum of Squares Within (SSW)

Sum of Squares Between (SSB)

Overall mean
of all 40
observations
("grand mean")

$$\bar{y}_{..} = \frac{\sum_{i=1}^4 \sum_{j=1}^{10} y_{ij}}{40}$$

$$\sum_{i=1}^4 10(\bar{y}_{i.} - \bar{y}_{..})^2 \longleftarrow$$

Sum of Squares Between
(SSB).

Total Sum of Squares (SST)

$$\sum_{i=1}^4 \sum_{j=1}^{10} (y_{ij} - \bar{\bar{y}}_{\bullet\bullet})^2$$

Total sum of squares(TSS).
Squared difference of every observation from the overall mean.

Decomposition of Variance (作业)

$$SSW + SSB = TSS$$

ANOVA Table

Source of variation	d.f.	Sum of squares	Mean Sum of Squares	F-statistic	p-value
Between (k groups)	k-1	SSB (sum of squared deviations of group means from grand mean)	SSB/k-1	$\frac{SSB / k - 1}{SSW / nk - k}$	Go to $F_{k-1, nk-k}$ chart
Within (n individuals per group)	nk-k	SSW (sum of squared deviations of observations from their group mean)	$s^2 = SSW / nk - k$		
Total variation	nk-1	TSS (sum of squared deviations of observations from grand mean)		$TSS = SSB + SSW$	

ANOVA=t-test

$$\begin{aligned}
 SSB &= n \sum_{i=1}^n (\bar{X}_n - (\frac{\bar{X}_n + \bar{Y}_n}{2}))^2 + n \sum_{i=1}^n (\bar{Y}_n - (\frac{\bar{X}_n + \bar{Y}_n}{2}))^2 = \\
 &= n \sum_{i=1}^n (\frac{\bar{X}_n}{2} - \frac{\bar{Y}_n}{2})^2 + n \sum_{i=1}^n (\frac{\bar{Y}_n}{2} - \frac{\bar{X}_n}{2})^2 \\
 &= n((\frac{\bar{X}_n}{2})^2 + (\frac{\bar{Y}_n}{2})^2 - 2\frac{\bar{X}_n * \bar{Y}_n}{2} + (\frac{\bar{Y}_n}{2})^2 + (\frac{\bar{X}_n}{2})^2 - 2\frac{\bar{X}_n * \bar{Y}_n}{2}) = \\
 &= n(\bar{X}_n^2 - 2\bar{X}_n * \bar{Y}_n + \bar{Y}_n^2) = n(\bar{X}_n - \bar{Y}_n)^2
 \end{aligned}$$

Source of variation	d.f.	Sum of squares	Mean Sum of Squares	F-statistic	p-value
Between (2 groups)	1	SSB (squared difference in means multiplied by n)	Squared difference in means times n	$\frac{n(\bar{X} - \bar{Y})^2}{s_w^2} = (\frac{(\bar{X} - \bar{Y})}{\sqrt{\frac{s_w^2}{n} + \frac{s_w^2}{n}}})^2 = (t_{2n-2})^2$	Go to $F_{1, 2n-2}$ Chart → notice values are just $(t_{2n-2})^2$
Within	2n-2	SSW equivalent to numerator of pooled variance	Pooled variance		
Total variation	2n-1	TSS			

Example

Treatment 1	Treatment 2	Treatment 3	Treatment 4
60 inches	50	48	47
67	52	49	67
42	43	50	54
67	67	55	67
56	67	56	68
62	59	61	65
64	67	61	65
59	64	60	56
72	63	59	60
71	65	64	65

Example

Step 1) calculate the sum of squares between groups:

Mean for group 1 = 62.0

Mean for group 2 = 59.7

Mean for group 3 = 56.3

Mean for group 4 = 61.4

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Grand mean= 59.85

$SSB = [(62-59.85)^2 + (59.7-59.85)^2 + (56.3-59.85)^2 + (61.4-59.85)^2] \times n \text{ per group}$
 $= 19.65 \times 10 = \mathbf{196.5}$

Example

Step 2) calculate the sum of squares within groups:

$(60-62)^2 + (67-62)^2 + (42-62)^2 + (67-62)^2 + (56-62)^2 + (62-62)^2 + (64-62)^2 + (59-62)^2 + (72-62)^2 + (71-62)^2 + (50-59.7)^2 + (52-59.7)^2 + (43-59.7)^2 + (67-59.7)^2 + (67-59.7)^2 + (69-59.7)^2 + \dots$ (sum of 40 squared deviations) = **2060.6**

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Step 3) Fill in the ANOVA table

<u>Source of variation</u>	<u>d.f.</u>	<u>Sum of squares</u>	Mean Sum of <u>Squares</u>	<u>F-statistic</u>	<u>p-value</u>
Between	3	196.5	65.5	1.14	.344
Within	36	2060.6	57.2	-	-
Total	39	2257.1	-	-	-

Step 3) Fill in the ANOVA table

<u>Source of variation</u>	<u>d.f.</u>	<u>Sum of squares</u>	Mean Sum of <u>Squares</u>	<u>F-statistic</u>	<u>p-value</u>
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Within	36	2060.6	57.2	-	-
Total	39	2257.1	-	-	-

Coefficient of Determination

$$R^2 = \frac{SSB}{SSB + SSW} = \frac{SSB}{SST}$$

The amount of variation in the outcome variable (dependent variable) that is explained by the predictor (independent variable).

Beyond one-way ANOVA

Often, you may want to test more than 1 treatment. ANOVA can accommodate more than 1 treatment or factor, so long as they are independent. Again, the variation partitions beautifully!

$$TSS = SSB1 + SSB2 + SSW$$

ANOVA summary

- A statistically significant ANOVA (F-test) only tells you that *at least* two of the groups differ, but not which ones differ.
- Determining *which* groups differ (when it's unclear) requires more sophisticated analyses to correct for the problem of multiple comparisons...